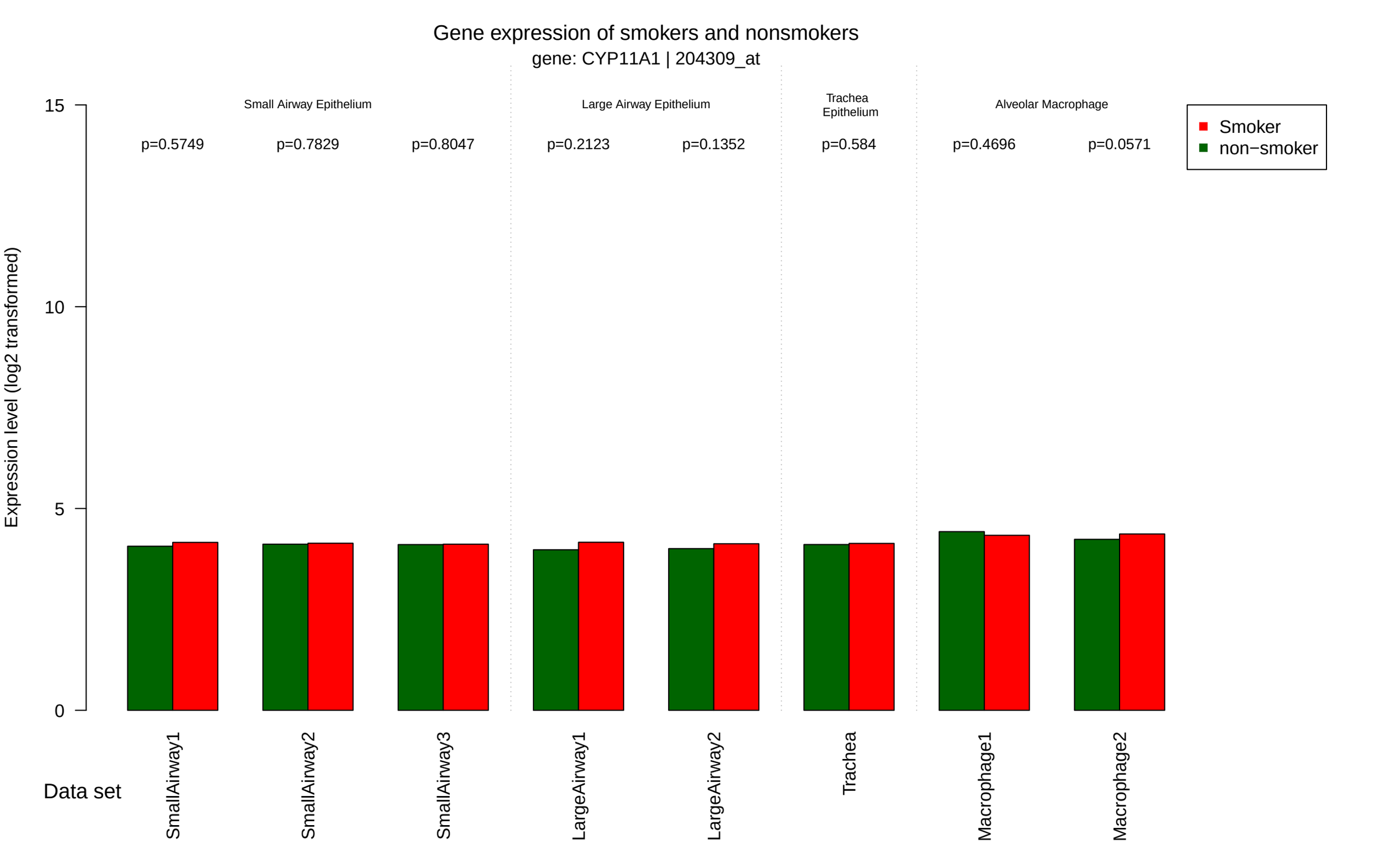
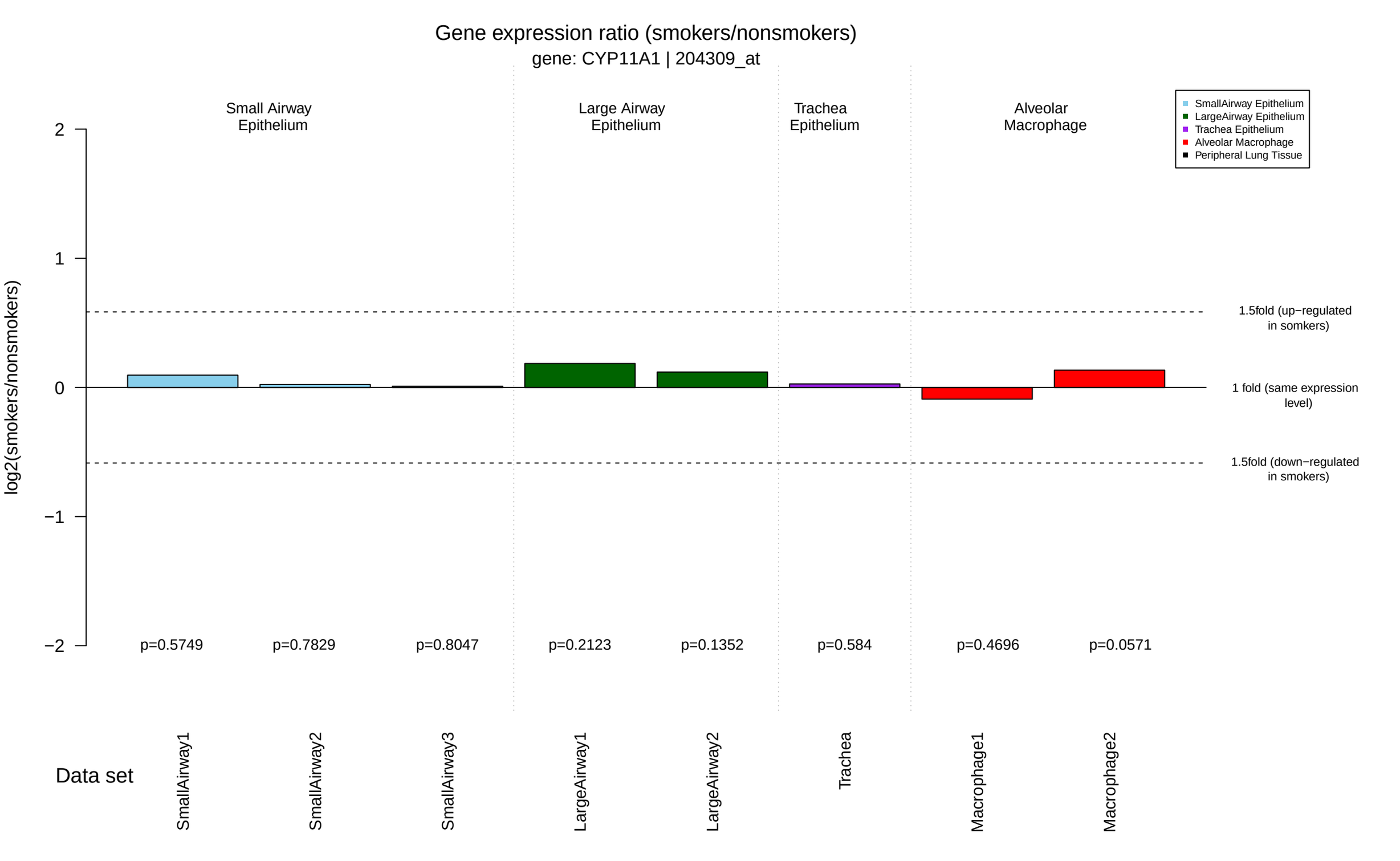
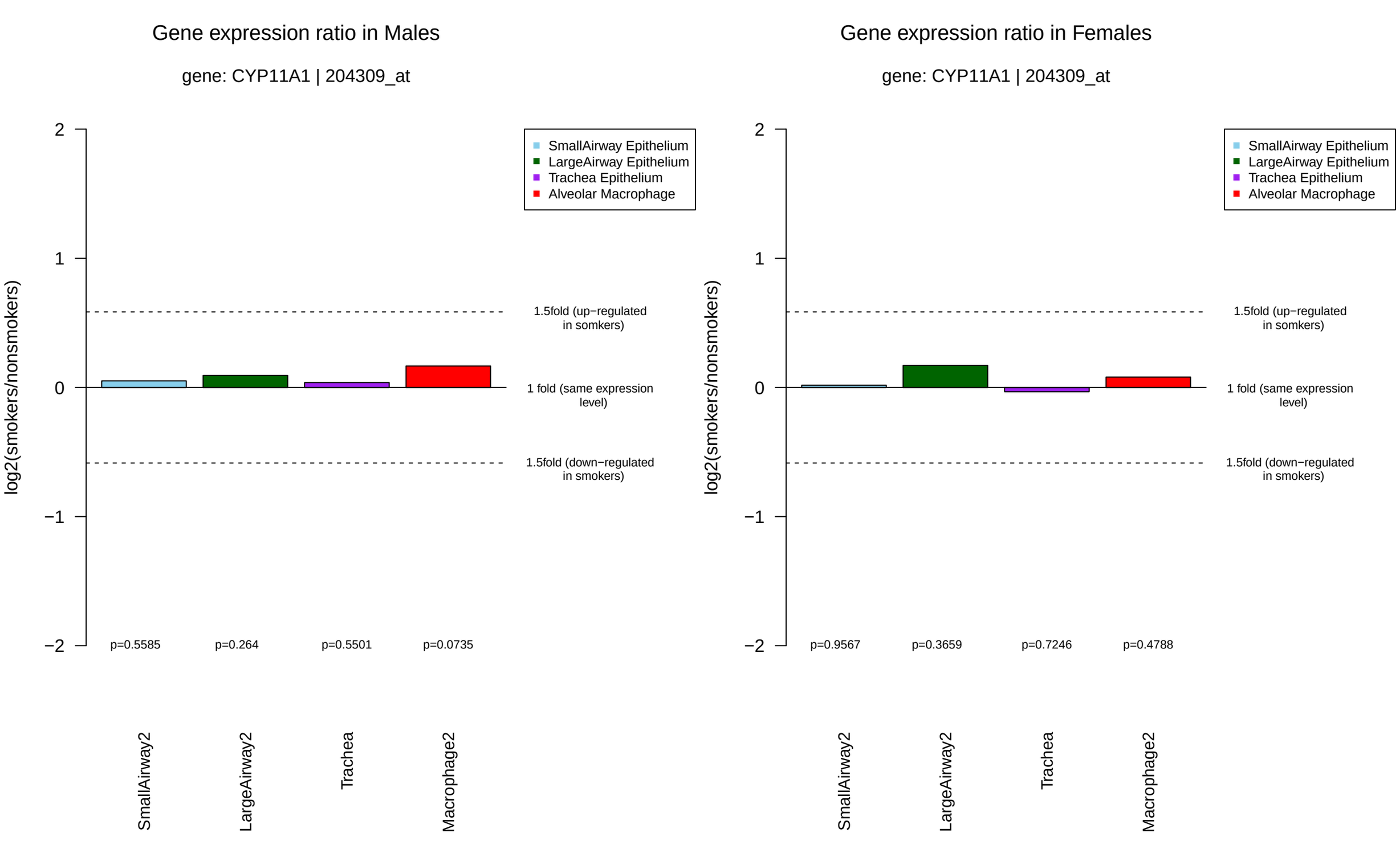
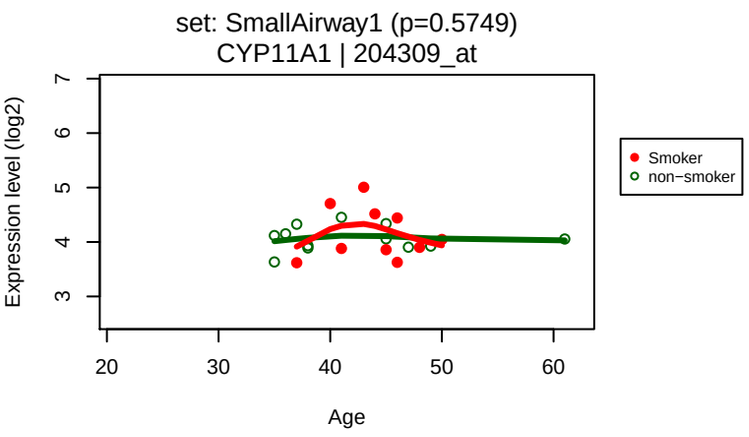
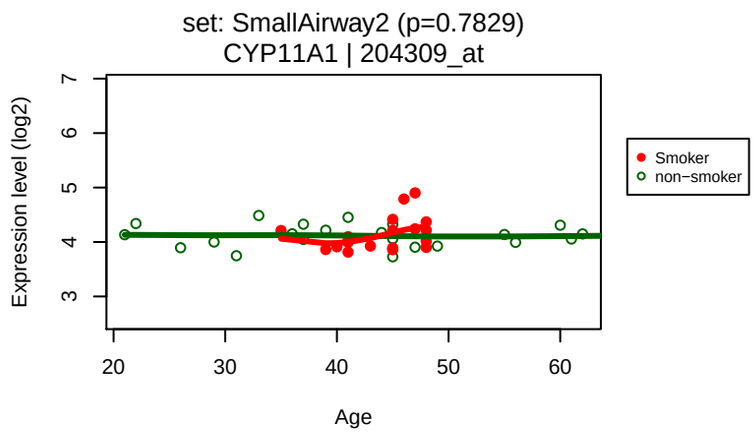
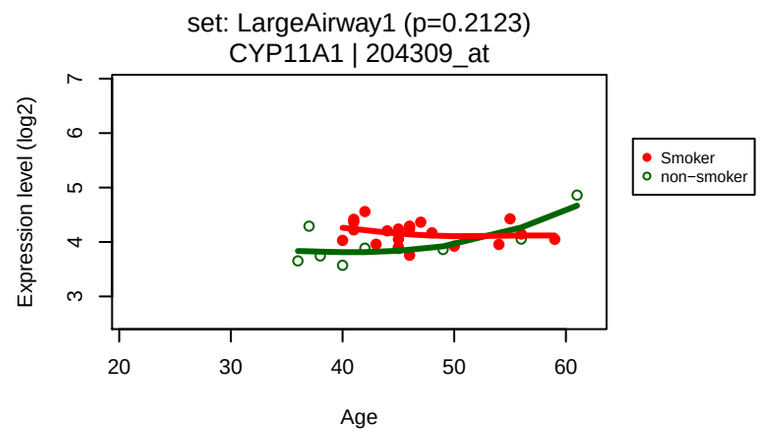

Gene expression of smokers and nonsmokers
gene: CYP11A1 | 204309_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5749
p=0.7829
p=0.8047
p=0.2123
p=0.1352
p=0.584
p=0.4696
p=0.0571
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

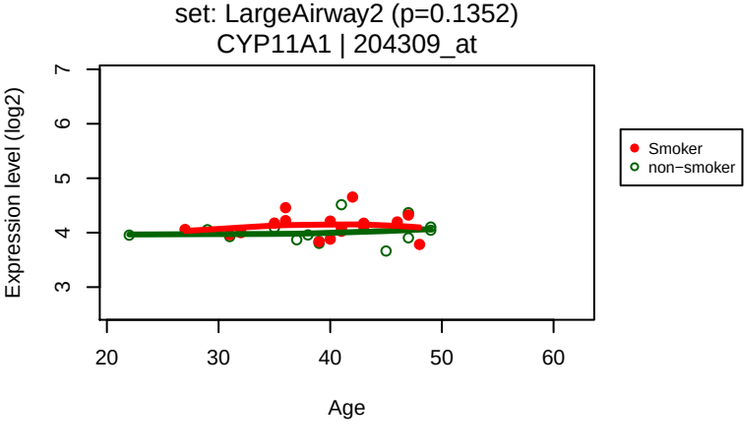

Gene expression ratio (smokers/nonsmokers)
gene: CYP11A1 | 204309_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5749
p=0.7829
p=0.8047
p=0.2123
p=0.1352
p=0.584
p=0.4696
p=0.0571
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

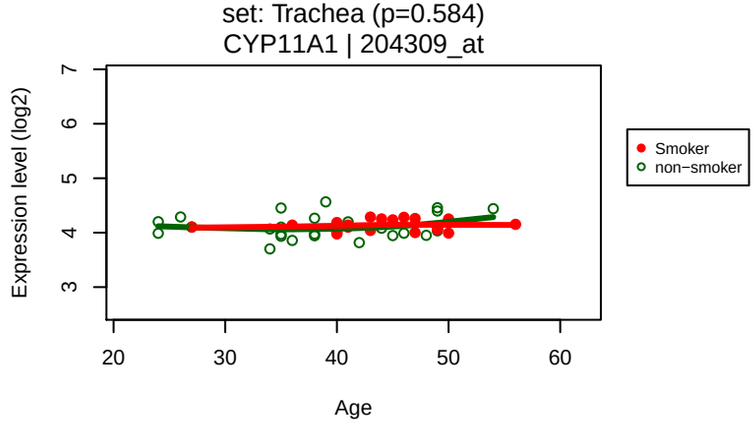

Gene expression ratio in Males
gene: CYP11A1 | 204309_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5585
p=0.264
p=0.5501
p=0.0735
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP11A1 | 204309_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9567
p=0.3659
p=0.7246
p=0.4788
SmallAirway2
LargeAirway2
Trachea
Macrophage2

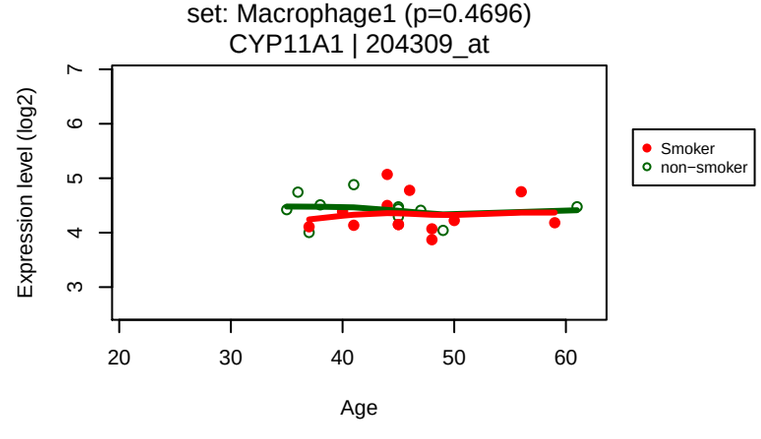

set: SmallAirway1 (p=0.5749)
CYP11A1 | 204309_at
Expression level (log2)
Age
Smoker
non−smoker

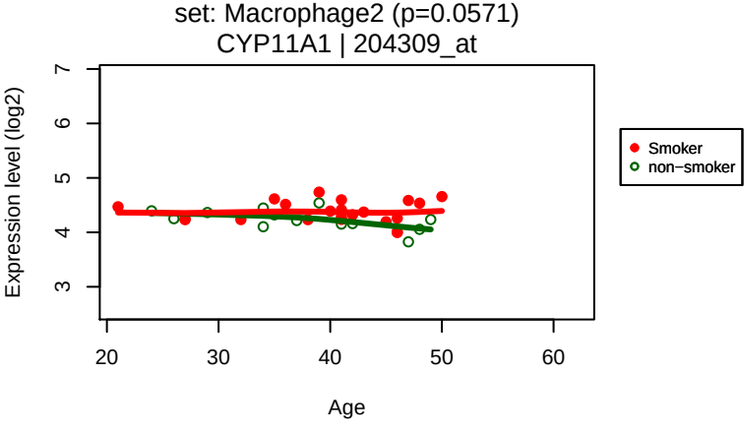

set: SmallAirway2 (p=0.7829)
CYP11A1 | 204309_at
Expression level (log2)
Age
Smoker
non−smoker

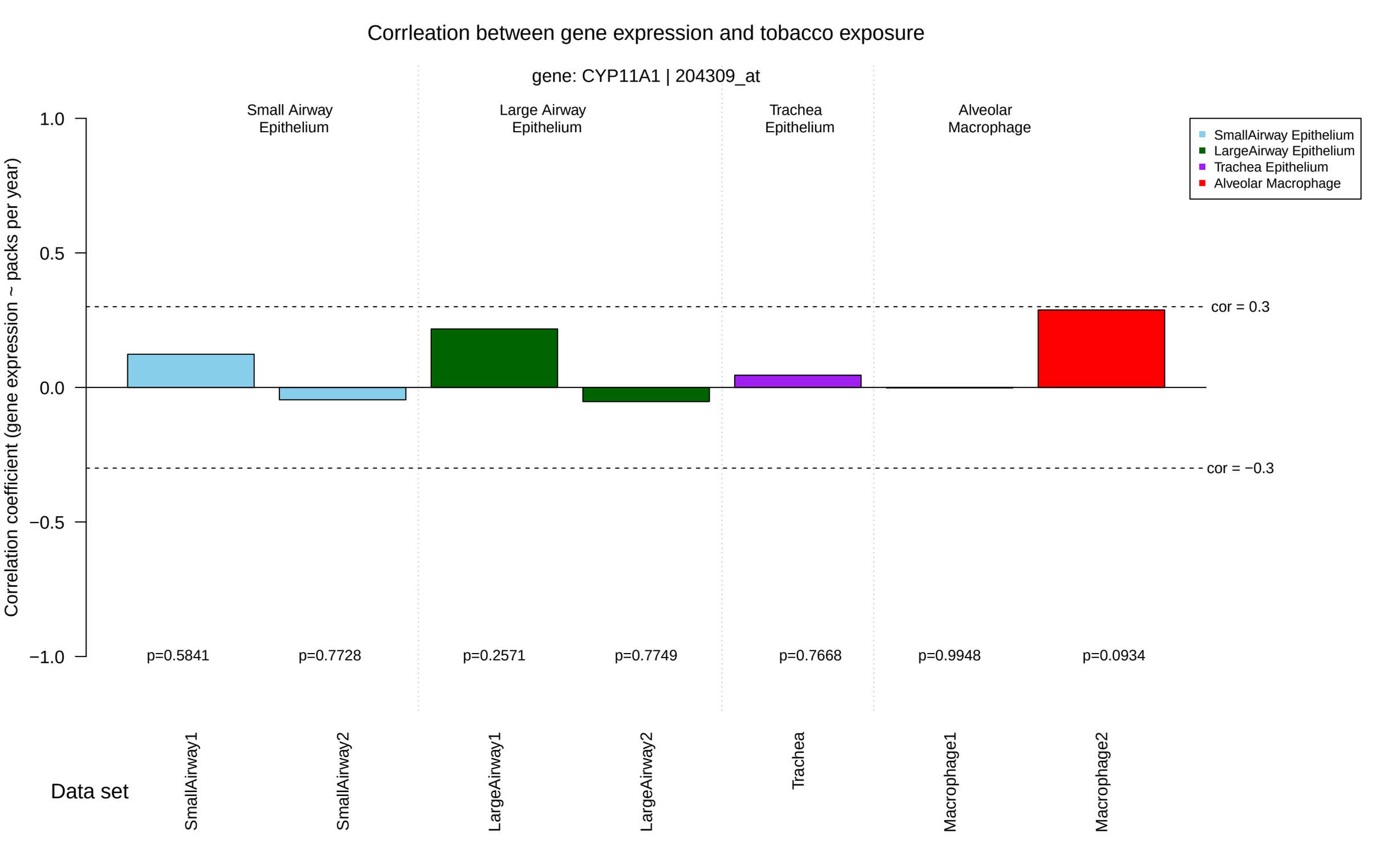

set: LargeAirway1 (p=0.2123)
CYP11A1 | 204309_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1352)
CYP11A1 | 204309_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.584)
CYP11A1 | 204309_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4696)
CYP11A1 | 204309_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0571)
CYP11A1 | 204309_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP11A1 | 204309_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5841
p=0.7728
p=0.2571
p=0.7749
p=0.7668
p=0.9948
p=0.0934
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2